

CSEC Mathematics Paper 02 – May/June 2026

Mark Scheme

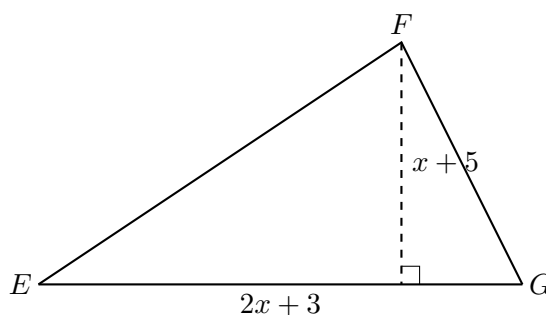
SECTION I

Question 1

- (a) (i) 0.367 to 1 significant figure is 0.4. 3.94 to 1 significant figure is 4. (1 mark)
- (a) (ii) Using the estimated values: $\frac{\sqrt{1000 \times 0.4}}{4}$. This simplifies to $\frac{\sqrt{400}}{4} = \frac{20}{4} = 5$. (2 marks)
- (b) The ratio of shares for Aiden, Brooke, and Caden is 3:2:5. The difference in parts between Aiden and Brooke is $3 - 2 = 1$ part. Since Aiden receives 12 more berries than Brooke, 1 part = 12 berries. The total number of parts is $3 + 2 + 5 = 10$. Total berries = $10 \times 12 = 120$. (2 marks)
- (c) (i) With a 15% reduction, Blake pays 85% of the original price. Original price is \$16.40. Amount paid = $\$16.40 \times 0.85 = \13.94 . (2 marks)
- (c) (ii) Jude paid \$27.20, which represents 85% of the original price. Original price = $\$27.20 \div 0.85 = \32.00 . (2 marks)

Question 2

- (a) (i) Substitute $p = 2$, $q = -1$, and $r = 1$ into $A = \frac{p-q}{2r}$. $A = \frac{2-(-1)}{2(1)} = \frac{3}{2} = 1.5$. (1 mark)
- (a) (ii) $A = \frac{p-q}{2r} \implies 2Ar = p - q \implies q = p - 2Ar$. (2 marks)
- (b) (i) Area of Triangle EFG = 50 cm^2 . The formula for area is $\frac{1}{2} \times \text{base} \times \text{height}$. Base EG = $2x + 3$ and height = $x + 5$.



$$\begin{aligned}\frac{1}{2}(2x + 3)(x + 5) &= 50 \\ (2x + 3)(x + 5) &= 100 \\ 2x^2 + 10x + 3x + 15 &= 100 \\ 2x^2 + 13x - 85 &= 0\end{aligned}$$

(3 marks)

- (b) (ii) Using the quadratic formula for $2x^2 + 13x - 85 = 0$:

$$x = \frac{-13 \pm \sqrt{13^2 - 4(2)(-85)}}{2(2)}$$

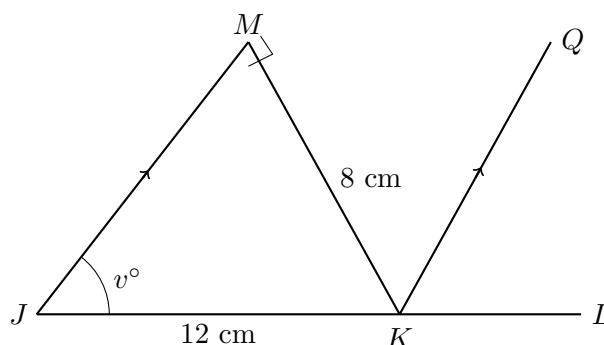
$$x = \frac{-13 \pm \sqrt{169 + 680}}{4} = \frac{-13 \pm \sqrt{849}}{4}$$

$$x = \frac{-13 \pm 29.1376}{4}$$

Since length must be positive, discard the negative root. $x = \frac{16.1376}{4} \approx 4.03$. (3 marks)

Question 3

- (a) (i) In right-angled triangle JMK, the hypotenuse JK = 12 cm and the opposite side to angle v° is MK = 8 cm. $\sin v^\circ = \frac{8}{12} = \frac{2}{3}$ or 0.667. (1 mark)



- (a) (ii) Using Pythagoras' theorem: $JM^2 + 8^2 = 12^2$. $JM^2 = 144 - 64 = 80$. $JM = \sqrt{80} \approx 8.94$ cm. (2 marks)
- (a) (iii) Since JM is parallel to KQ, $\angle MJK = \angle QKL = v^\circ$ (corresponding angles). $\angle MKL = 180^\circ - \angle JKM$. Alternatively, find angle JKM using $\cos(\angle JKM) = \frac{8}{12}$, so $\angle JKM \approx 48.2^\circ$. Thus, $\angle MKL = 180^\circ - 48.2^\circ = 131.8^\circ$. (2 marks)
- (b) Construction criteria:
 1. Draw straight line WX = 9 cm.
 2. Construct a 60° angle at X.
 3. Measure 5 cm along the constructed 60° ray to establish point Y.
 4. Construct a line parallel to WX passing through Y.
 5. Measure 3 cm along this parallel line to establish point Z.
 6. Draw a line to connect Z and W.

(4 marks)

Question 4

- (a) (i) Midpoint M of A(-3, -4) and B(2, 6): $\left(\frac{-3+2}{2}, \frac{-4+6}{2}\right) = (-0.5, 1)$. (2 marks)
- (a) (ii) Gradient $m = \frac{6-(-4)}{2-(-3)} = \frac{10}{5} = 2$. (2 marks)

- (a) (iii) Perpendicular bisector gradient is the negative reciprocal: $m_{\perp} = -0.5$. Using point M(-0.5, 1): $y - 1 = -0.5(x - (-0.5))$. This simplifies to $y = -0.5x + 0.75$ or $2y + x = 1.5$. (3 marks)
- (b) Gradient of AZ using A(-3, -4) and Z(-5, -8): $\frac{-8 - (-4)}{-5 - (-3)} = \frac{-4}{-2} = 2$. Since the gradient of AZ equals the gradient of AB and they share the common point A, the points A, B, and Z are collinear. (2 marks)

Question 5

- (a) Total number of matches = $3 + 10 + k + 15 + 4 + 3 = 35 + k$. Total goals = $(0 \times 3) + (1 \times 10) + (2 \times k) + (3 \times 15) + (4 \times 4) + (5 \times 3) = 86 + 2k$. Mean = $\frac{86+2k}{35+k} = 2.2$. Solving yields $86 + 2k = 77 + 2.2k \implies 9 = 0.2k \implies k = 45$. (3 marks)
- (b) (i) From the graph, at 42 seconds, the cumulative frequency is approximately 15. The number of students taking at least 42 seconds is $60 - 15 = 45$. Probability = $\frac{45}{60} = 0.75$. (2 marks)
- (b) (ii) Lower Quartile (Q_1) is the 15th value, reading across from 15 gives approximately 42 seconds. Upper Quartile (Q_3) is the 45th value, reading across from 45 gives approximately 64 seconds. SIQR = $\frac{64-42}{2} = \frac{22}{2} = 11$ seconds. (2 marks)
- (b) (iii) For interval 41-60: Frequency = $40 - 13 = 27$; Midpoint = 50.5; Frequency $\times$ Midpoint (fx) = 1363.5.
For interval 61-80: Frequency = $57 - 40 = 17$; Midpoint = 70.5; Frequency $\times$ Midpoint (fx) = 1198.5. (2 marks)

Question 6

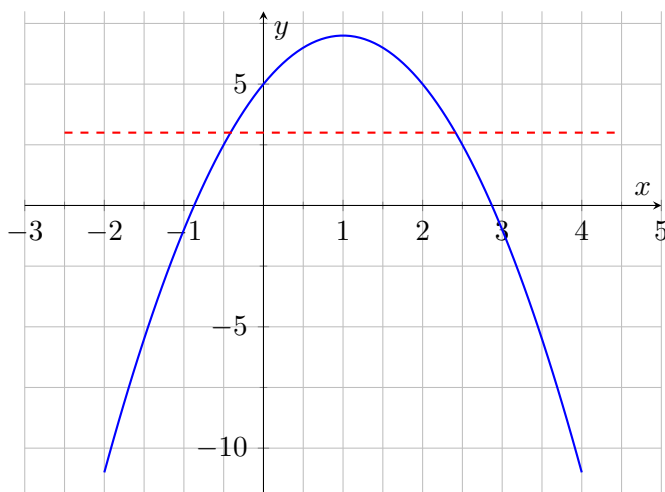
- (a) (i) Ratio of 175 g flour : 125 g butter : 75 g sugar. Dividing by 25 yields 7 : 5 : 3. (2 marks)
- (a) (ii) Multiplier for 52 cookies is $\frac{52}{20} = 2.6$. Flour = $175 \times 2.6 = 455$ g. Butter = $125 \times 2.6 = 325$ g. Sugar = $75 \times 2.6 = 195$ g. (3 marks)
- (b) One pack requires 12 cookies. Total flour available = 12.65 kg = 12650 g. Flour per single cookie = $\frac{175}{20} = 8.75$ g. Flour per pack = $8.75 \times 12 = 105$ g. Maximum packs = $\frac{12650}{105} \approx 120.47$, therefore 120 packs. Flour used = $120 \times 105 = 12600$ g. Amount left over = $12650 - 12600 = 50$ g. (4 marks)

Question 7

- (a) (i) Row 9, Column 1 follows the sequence multiples of 7: $7 \times 9 = 63$. (1 mark)
- (a) (ii) The middle column represents $7r + 2 = 653$, giving row $r = 93$. Column 1 = $7(93) = 651$. Column 3 = $651 + 4 = 655$. (2 marks)
- (a) (iii) Dividing 942 by 7 gives 134 with a remainder of 4. Row number is 134. Explanation: Since the numbers follow sequences of $7n$, $7n + 2$, and $7n + 4$, and 942 leaves a remainder of 4 when divided by 7, it sits in Column 3 of the 134th row. (2 marks)
- (b) (i) $E = 2n - 1$. $S = 4n - 1$. $D = 2n + 1$. (3 marks)
- (b) (ii) $D - E = (2n + 1) - (2n - 1)$. Distributing the negative gives $2n + 1 - 2n + 1 = 2$. Validated for all values of n . (2 marks)

Question 8

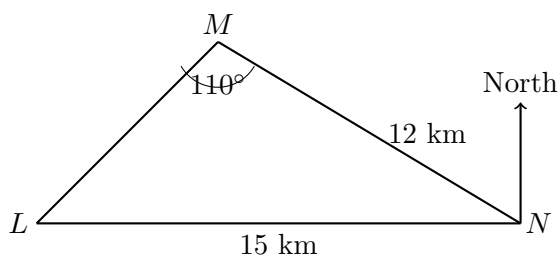
- (a) $f(x) = -2x^2 + 4x + 5 = -2(x^2 - 2x) + 5$. Completing the square: $-2(x^2 - 2x + 1 - 1) + 5 = -2(x - 1)^2 + 2 + 5 = -2(x - 1)^2 + 7$. Therefore, $a = -2$, $h = -1$, $k = 7$. (2 marks)
- (b) (i) Missing values: When $x = 1$, $y = 5 + 4(1) - 2(1)^2 = 7$. When $x = 2$, $y = 5 + 4(2) - 2(2)^2 = 5$. When $x = 3$, $y = 5 + 4(3) - 2(3)^2 = -1$. (2 marks)
- (b) (ii) Graph coordinates: Plot points $(-2, -11)$, $(-1, -1)$, $(0, 5)$, $(1, 7)$, $(2, 5)$, $(3, -1)$, $(4, -11)$ on the provided axes and connect with a smooth parabolic curve. (5 marks)



- (c) Draw the horizontal line $y = 3$ on the graph. The x-values at the points of intersection with the curve represent the solutions to $5 + 4x - 2x^2 = 3$. Accurate readouts from the graph will yield $x \approx -0.4$ and $x \approx 2.4$. (3 marks)

Question 9

- (a) (i) Vertices of H map to N by the rule $(x, y) \rightarrow (-y, x)$. This defines a rotation of 90° anti-clockwise about the origin $(0, 0)$. (3 marks)
- (a) (ii) Vertices of H map to P by adding 2 to the x-coordinate and subtracting 5 from the y-coordinate. This defines a translation by the vector $\begin{pmatrix} 2 \\ -5 \end{pmatrix}$. (2 marks)
- (b) (i) M vertices determined by applying scale factor 2 from centre $(0, -1)$ to H. Vertices of M should be plotted at $(4, 7)$, $(4, 3)$, $(6, 1)$, and $(8, 3)$. (2 marks)
- (b) (ii) L vertices determined by reflecting H across the line $y = -x$, applying the rule $(x, y) \rightarrow (-y, -x)$. Vertices of L should be plotted at $(-3, -2)$, $(-1, -2)$, $(0, -3)$, and $(-1, -4)$. (2 marks)
- (c) (i) Since L is due west of N, its bearing from N is 270° . The bearing of M from N is 291° . $\angle MNL = 291^\circ - 270^\circ = 21^\circ$. (1 mark)



- (c) (ii) Using the Cosine rule on $\triangle LMN$ with $MN = 12$, $LN = 15$, and $\angle MNL = 21^\circ$:

$$LM^2 = 15^2 + 12^2 - 2(15)(12)\cos(21^\circ)$$

$$LM^2 = 225 + 144 - 360(0.9336) = 369 - 336.1 = 32.9$$

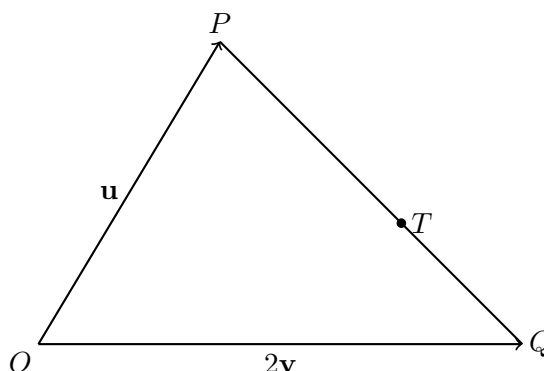
$$LM = \sqrt{32.9} \approx 5.74 \text{ km}$$

(2 marks)

Question 10

- (a) (i) $\vec{PQ} = \vec{PO} + \vec{OQ} = -\mathbf{u} + 2\mathbf{v}$.

(1 mark)



- (a) (ii) $\vec{OT} = \vec{OP} + \vec{PT}$. Since $PT : TQ = 3 : 2$, $PT = \frac{3}{5}\vec{PQ}$.

$$\vec{OT} = \mathbf{u} + \frac{3}{5}(-\mathbf{u} + 2\mathbf{v}) = \mathbf{u} - \frac{3}{5}\mathbf{u} + \frac{6}{5}\mathbf{v} = \frac{2}{5}\mathbf{u} + \frac{6}{5}\mathbf{v}$$

(2 marks)

- (b) (i)

a) Enlargement matrix: $\begin{pmatrix} 8 & 0 \\ 0 & 8 \end{pmatrix}$.

b) Reflection matrix: $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

c) Combined: $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 8 & 0 \\ 0 & 8 \end{pmatrix} = \begin{pmatrix} 8 & 0 \\ 0 & -8 \end{pmatrix}$.

(4 marks)

- (b) (ii)

a) $RQ = \begin{pmatrix} 6 & 1 \\ 4 & 2 \end{pmatrix} \begin{pmatrix} k & 1 \\ c & -6 \end{pmatrix} = \begin{pmatrix} 6k + c & 6 - 6 \\ 4k + 2c & 4 - 12 \end{pmatrix} = \begin{pmatrix} 6k + c & 0 \\ 4k + 2c & -8 \end{pmatrix}$.

(2 marks)

- b) Equating elements of RQ with the matrix from (b)(i)(c):

$$6k + c = 8 \quad \text{and} \quad 4k + 2c = 0$$

From the second equation, $c = -2k$. Substitute into the first:

$$6k + (-2k) = 8 \implies 4k = 8 \implies k = 2$$

Substitute $k = 2$ into $c = -2k$: $c = -2(2) = -4$.

(3 marks)